

Cloud Computing

Mod 2: Evolution of Cloud Computing and its models

Cloud computing is not an isolated development. Cloud technology has matured over the years with constant advancements in the field of computing. Several decades of research, particularly in the domain of parallel and distributed computing, have paved the way for cloud computing. Cloud computing has evolved significantly, transforming how businesses and individuals access technology resources.

Early Foundations (1960s–1980s):

- Distributed computing emerged, enabling resource sharing.
- Grid computing in the 1980s allowed task distribution across networks for efficiency.

Emergence of Cloud Models (1990s–2006):

- Virtualization enabled multiple virtual machines on a single server.
- Salesforce (1999) pioneered Software as a Service (SaaS).
- AWS launched Elastic Compute Cloud (EC2) in 2006, revolutionizing scalable, on-demand computing.

Growth and Adoption:

- Google Cloud and Microsoft Azure popularized the pay-as-you-go model.
- Businesses adopted cloud services for cost reduction and agility.

Technological Advancements:

- Platform as a Service (PaaS) simplified app development.
- Cloud-native technologies like containers and serverless computing improved scalability.

Future Trends:

- Edge computing reduces latency by processing data closer to its source.
- AI integration enhances data analysis.
- Security, data privacy, and regulatory compliance remain critical concerns.

Cluster Computing: Creation of Resource Pool

The concept of clustering appeared as the next step of evolution in the field of computing. Computing clusters are made of multiple nodes (computers) connected via network which perform similar tasks. Thus, execution of a task can be faster as it can be distributed and executed in parallel across multiple machines inside a cluster. All the nodes of a cluster together give the impression of a single system. Each node in a simple cluster is set to perform the same task or same set of tasks. The idea was to create a cluster (or group) of homogeneous (similar type) computer systems performing similar functionalities. In

each cluster, one computer is assigned the job of controlling the cluster. That particular computer (or node) is known as cluster head. The head's responsibility in such a simple cluster is to divide and distribute jobs among different nodes in that cluster when matching computing tasks appear. In an actual cluster computing system, multiple clusters (built to perform different types of functionalities) are linked together through a LAN. In such a computing environment, when a particular job appears, the cluster head divides and distributes it among matching clusters (designated for those jobs) for faster execution. The distribution and assignment of job depends on the nature of the job. This way, the cluster head starts to utilize the resources in a clustered computing environment. Computers are clustered together to achieve reliability, greater processing power and produce supercomputer-like performance. The cluster computing model emerged as a result of progress in multiple technologies and their convergence. Availability of low cost microprocessors, high speed network communication and emergence of software to manage cluster distribution made cluster computing possible. In this computing model, a set of computers were reserved to handle specific types of tasks to make the system more reliable. If any node fails, other nodes in the cluster can handle the load. This was the idea behind the resource pool that technologists were trying to implement, and clustering of nodes opened the path for creating pools of resources. With this implementation, reliability was achieved through redundancy.

Grid Computing: Decentralization of Control

Cluster architecture led to more powerful and reliable computing systems by creating resource pools, but it raised concerns regarding its dependency on the cluster head. Performance of such systems was largely dependent on the efficiency and accessibility of cluster heads. Existence of cluster heads raises the possibility for a single point of failure too. An advanced computing model was required to eliminate the cluster head problem of the cluster computing model. In the process of finding a solution to this problem, technologists came up with an idea where each node belonging to a cluster would have the same priority. It was required that all of them could perform similar functions and no particular node had to be assigned the role of 'head' among them. This new architecture was introduced in the early 1990s, and was called a grid. Other than decentralization of control, grid computing introduced another important system feature. The computing environment could now be built with heterogeneous computing systems, that is, systems with diverse hardware configurations. This was a huge benefit as resource pools could then grow by accommodating heterogeneous systems. The other major challenge that emerged as a stumbling block before setting up a large distributed computing environment spread around the world was establishing cooperation among systems of different administrative domains. For multiple administrative domains, the challenge was related to management of resources and security implementations of systems. Initially, grid computing was used particularly by the science and engineering community. But when enterprise applications adopted the grid computing model, it became necessary to address the need for uniform security implementation and other resource management policies. Technologists focused on bringing

about uniformity in these areas, during that period. Once it was done, the evolution moved on to the next level.

With the grid computing era, computing systems reached a stage where they had acquired the following important characteristics.

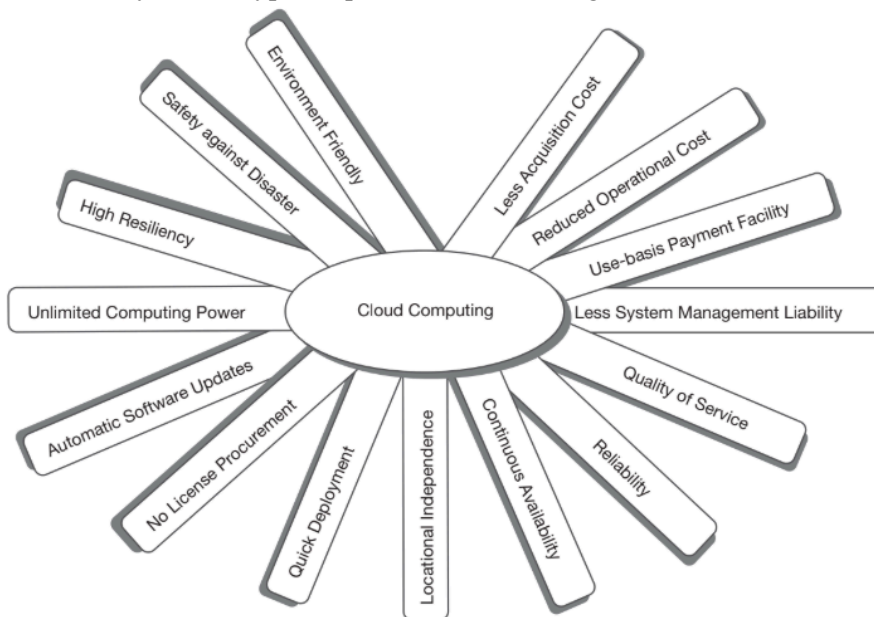
- Large scale: Grid concept promises to deal with computing resources going up to millions.
- Geographical distribution: Computing resources could be located at distant places.
- Heterogeneity: Computing grid could accommodate resources having different hardware characteristics and using various types of software.
- Resource coordination: Resources in a grid could coordinate among themselves to generate aggregated computing capabilities.
- Pervasive access: Resource failure could be handled by granting access to other available resources.
- Unlimited resource addition (scaling): Being a distributed computing model, it allows easy growth of system capacity by adding more resources into an existing system.

<i>Characteristic</i>	<i>Cluster Computing</i>	<i>Grid Computing</i>
Ownership	Single	Multiple
System management	Centralized	Decentralized
User management	Centralized	Decentralized
Resource Pool Creation	Yes	Yes
Resource node type	Homogeneous	Heterogeneous
Resource scheduling	Centralized	Decentralized
Network connection type*	LAN	LAN/MAN (Metropolitan area network)/WAN
System architecture	Single system image	Autonomous, independent nodes
Coupling	Tight	Loose
Fault tolerance Capability	Low	High
Scaling Architecture	Scalable	Scalable, in bigger capacity
Real-time Scaling*	No	No

Benefits of Cloud computing

- Cost Savings Cloud computing can lead to cost savings by reducing the need for costly hardware and software⁴. The pay-as-you-go model of cloud computing eliminates the need for large upfront investments, which is especially beneficial for startups and SMEs.
- Scalability and Flexibility Cloud computing provides unparalleled scalability, allowing businesses to easily scale their resources up or down based on demand. Cloud capacity and resources can rapidly expand to meet user demands.
- Access to New Technologies Organizations gain early and instant access to the IT industry's latest technologies, such as automatically updated applications, machine learning, and AI, through large cloud providers.
- Rapid Deployment Cloud computing allows organizations to install applications and services quickly, reducing time-to-market.
- Improved Collaboration Cloud computing centralizes data and applications, making them accessible from a single platform and improving collaboration by providing team members with access to the same data and tools, regardless of location.
- Automation: Cloud computing makes it easier for businesses to automate tasks, such as provisioning servers or configuring networks, which can significantly reduce the amount of time needed to complete these tasks manually.

- Data Backup and Restore Cloud-based solutions offer vast data storage, which is beneficial for backup and restore operations.
- Enhanced Data Security: Cloud providers use sophisticated security measures, such as encryption and multi-factor authentication, to protect data from unauthorized access or malicious attacks.
- Analytics Public: cloud services can perform analytics on high volumes and accommodate a variety of data types to present business insights.



Challenges of Cloud Computing

- Runaway Costs: Complex cloud costs and pricing models can make it difficult for organizations to keep track of IT spending.
- Security and Privacy: Concerns Storing sensitive data on distant servers exposes businesses to potential threats, and a security breach could have severe effects, including economic loss and reputational harm.
- Scarce Cloud Expertise: Companies struggle to hire and retain staff with expertise in building and managing modern cloud applications.
- Limited Controls: Public cloud users have limited control over their IT stack because the provider can decide when and how to manage configurations.
- Downtime and Service Reliability" Cloud computing relies on internet connectivity, and any disruption can result in downtime and loss of access to vital data and packages.
- Compliance Issues" There can be compliance issues.
- Vendor Lock-in: There can be vendor lock-in.
- Network Latency Issue: There can be network latency issues.

<i>Business Challenges</i>	<i>Cloud Computing Solutions</i>
Budget	It provides flexibility by maximizing utilization of available budget with the scope for future growth
Business	It reduces time to market and provides scope for rapid business growth.
Mobility of users	It enables anywhere access from a variety of devices.
Flexibility	The agile nature of cloud computing enables easier and quick changes.
Scope of Growth	The scalable (or elastic) architecture enables rapid growth.
Availability	It provides almost 24 × 7 availability round the clock.
Recovery	It integrates reliable backup and disaster recovery mechanisms.

Cluster Computing	Grid Computing	Cloud Computing
A cluster is normally formed with computers of a single location, otherwise the system becomes complex.	Grid is inherently more distributed by its nature. The computers need not to be in the same geographical location.	It allows total distribution of resources like the grids. Hardware resources are maintained in multiple data centers spread across the globe.
Computation job takes place in one administrative domain owned by a single party.	Computation could occur over many administrative domains owned by multiple parties as connected together.	Computing resources of a cloud is usually owned by a single party. But multiple administrative domains can be combined together to perform the job.
In a cluster, all computing nodes should have similar hardware systems. That is, the system should be homogeneous in nature.	It can be heterogeneous in nature. The computers that are part of a grid can be made of different hardware architectures.	It can use heterogeneous collection of commodity hardware.
It features the centralized task management and scheduling system.	It features the distributed task management and decentralized scheduling.	It features the decentralized task management with more dynamic computing infrastructure.
Resources are generally pre-reserved for specific type of task.	Resources are generally pre-reserved for specific type of task.	Resources are not pre-reserved for specific task. Resource utilization is mainly demand-driven.
System is not dynamic in nature. Application mobility is not possible.	System is not dynamic in nature. Application mobility is not possible.	It is a dynamic system. Mobility of application is an inherent feature in this system.
One whole cluster behaves like a single system. As resources are managed by centralized resource manager, the individual computers can not be operated as separate computers.	Every node is autonomous that is, it has its own resource manager and behaves like an independent entity. So, each computer can be operated independently as distinct computer.	There is no concept of directly accessing any particular physical computing nodes. Underlying computing infrastructure remains hidden from the users.

NIST Models

Cloud Computing Definition: NIST defines cloud computing as a model that enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This definition is foundational for understanding the operational aspects of cloud services.

Essential Characteristics:

- **On-Demand Self-Service:** Users can provision computing capabilities automatically without requiring human interaction with each service provider.
- **Broad Network Access:** Services are available over the network and accessed through standard mechanisms that promote use across various platforms (e.g., mobile phones, tablets).
- **Resource Pooling:** The provider's resources are pooled to serve multiple consumers using a multi-tenant model, dynamically assigning resources according to demand.
- **Rapid Elasticity:** Capabilities can be elastically provisioned and released to scale rapidly outward and inward commensurate with demand.
- **Measured Service:** Resource usage is monitored, controlled, and reported providing transparency for both the provider and consumer

Table 4.1 Comparison of characteristics between cluster, grid and cloud computing models

	<i>Cluster Model</i>	<i>Grid Model</i>	<i>Cloud Model</i>
Resource Pooling	Yes	Yes	Yes
Broad Network Access	Yes	Yes	Yes
Measured Services	No	Yes	Yes
Rapid Elasticity	No	No	Yes
On-demand Self-service	No	No	Yes

Service Models:

The highest-level categorization of cloud services as based on the type of computing capability that is provided. Any given cloud service may be categorized as one of three service models, namely Software as a Service (SaaS), Platform as a Service (PaaS), or Infrastructure as a Service (IaaS).

IaaS: Infrastructure as a Service (IaaS) is a means of delivering computing infrastructure as on-demand services. It is one of the three fundamental cloud service models. The user purchases servers, software data center space, or network equipment and rents those resources through a fully outsourced, on-demand service model. It allows dynamic scaling and the resources are distributed as a service. It generally includes multiple users on a single piece of hardware. It totally depends upon the customer to choose its resources wisely and as per need. Also, it provides billing management.

Characteristics of IaaS (Infrastructure as a Service)

- IaaS is like renting virtual computers and storage space in the cloud.
- You have control over the operating systems, applications, and development frameworks.
- Scaling resources up or down is easy based on your needs.

Example of IaaS (Infrastructure as a Service)

- Amazon Web Services, Microsoft Azure, Google Compute Engine, Digital Ocean

PaaS: Platform as a Service (PaaS) is a cloud delivery model for applications composed of services managed by a third party. It provides elastic scaling of your application which allows developers to build applications and services over the internet and the deployment models include public, private and hybrid. Basically, it is a service where a third-party provider provides both software and hardware tools to the cloud computing. The tools which are provided are used by developers. PaaS is also known as Application PaaS. It helps us to organize and maintain useful applications and services. It has a well-equipped management system and is less expensive compared to IaaS.

Characteristics of PaaS (Platform as a Service)

- PaaS is like a toolkit for developers to build and deploy applications without worrying about infrastructure.
- Provides pre-built tools, libraries, and development environments.

- Developers focus on building and managing applications, while the provider handles infrastructure management.
- It speeds up the development process and allows for easy collaboration among developers.

Examples of PaaS (Platform as a Service)

- AWS Lambda, Google App Engine, Google Cloud, IBM Cloud

SaaS: Software as a Service (SaaS) allows users to run existing online applications and it is a model software that is deployed as a hosting service and is accessed over the internet or software delivery model during which software and its associated data are hosted centrally and accessed using their client, usually an online browser over the web. SaaS services are used for the development and deployment of modern applications. It allows software and its functions to be accessed from anywhere with a good internet connection device and a browser. An application is hosted centrally and also provides access to multiple users across various locations via the internet.

Characteristics of SaaS (Software as a Service)

- Applications are ready to use, and updates and maintenance are handled by the provider.
- You access the software through a web browser or app, usually paying a subscription fee.
- It's convenient and requires minimal technical expertise, ideal for non-technical users.

Example of SaaS (Software as a Service)

- Salesforce, Google Workspace apps, Microsoft 365, Trello, Zoom, Slack, Adobe Creative Cloud

Variables	SaaS	PaaS	IaaS
Full form	Software as a service	Platform as a service	Infrastructure as a service
Used for	It is used for cloud computing software development	It is used for application development in cloud computing	It is used for any cloud computing development
Type of users	Any users who use it	It is used by software and application developers	Network architects use it
Access	SaaS gives access to any software that you want to use	It provides the necessary tools for development	You can store data and also use virtual machines
Technical know-how	If you want to use SaaS, you do not have to be a technical expert	You need to have a piece of proper knowledge of developing applications	You need to have excellent technical knowledge
Popularity	Most consumers and businesses focus on SaaS	PaaS is primarily popular among application developers	IaaS is popular among researchers and developers
Cost	The cost of using SaaS is minimum as you use the software according to your requirements	The cost of using PaaS is low	The cost of using IaaS is the highest

Types of Cloud Computing Deployment Models

The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose. The location of the servers you're utilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself.

Public Cloud: The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. For example, Google App Engine etc.

Advantages of the Public Cloud Model

- Minimal Investment: Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- No setup cost: The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- Infrastructure Management is not required: Using the public cloud does not necessitate infrastructure management.
- No maintenance: The maintenance work is done by the service provider (not users).
- Dynamic Scalability: To fulfill your company's needs, on-demand resources are accessible.

Disadvantages of the Public Cloud Model

- Less secure: Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- Low customization: It is accessed by many public so it can't be customized according to personal requirements.

Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between private and public clouds is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.

Advantages of the Private Cloud Model

- Better Control: You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- Data Security and Privacy: It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- Supports Legacy Systems: This approach is designed to work with legacy systems that are unable to access the public cloud.
- Customization: Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

Disadvantages of the Private Cloud Model

- Less scalable: Private clouds are scaled within a certain range as there is less number of clients.
- Costly: Private clouds are more costly as they provide personalized facilities.

Hybrid Cloud: By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.

Advantages of the Hybrid Cloud Model

- Flexibility and control: Businesses with more flexibility can design personalized solutions that meet their particular needs.
- Cost: Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- Security: Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages of the Hybrid Cloud Model

- Difficult to manage: Hybrid clouds are difficult to manage as it is a combination of both public and private clouds. So, it is complex.
- Slow data transmission: Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Community Cloud: It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.

Advantages of the Community Cloud Model

- Cost Effective: It is cost-effective because the cloud is shared by multiple organizations or communities.
- Security: Community cloud provides better security.
- Shared resources: It allows you to share resources, infrastructure, etc. with multiple organizations.
- Collaboration and data sharing: It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model

- Limited Scalability: Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- Rigid in customization: As the data and resources are shared among different organizations according to their mutual interests, if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

Choosing an Appropriate Deployment Model

The choice of appropriate cloud deployment depends on several factors. It largely depends on the business needs and also on the size and IT maturity of consumer organization. Consumers should analyze the pros and cons of on-premises and off-premises deployment options and must be careful before selecting a cloud deployment model as different deployments are best fitted for serving different requirements. Business need and security need (especially that of data) are two important factors which play a significant role in the decision making.

For general users, any reputed public cloud service is a good option. The issue regarding the appropriate choice of cloud deployment mainly stands before organizations (and communities also). Any reputed public cloud can be a choice for them but private (or community) deployment becomes the likely option when concern is about the privacy of sensitive or importance of some vital business-related data. Even in the case of setting up an in-house cloud, an organization (or community) must consider the capability of their in-house technical team; otherwise they have the choice of outsourcing the (private or community cloud) service. While outsourcing, the expertise or reputation of the service provider has to be verified.

Budget is another important issue. The cost of migration into the cloud and the total cost of ownership have to be considered before selecting a deployment. Generally, for a critical application that has security issues, a private or hybrid cloud model may suit well. On the other hand, for a general application, the public cloud may serve the purpose better. It is also critical to understand the business goals of consumer organization based on many functional as well as other non-functional requirements. For instance, the cost of computing and consumer's control over the computing environment directly change with the choice of deployment model. Following section discusses two such aspects. To select the appropriate cloud deployment model, one will have to assess the business needs, strengths and weaknesses of a model, privacy/security requirements and goals of the organization.

FEATURE	PUBLIC CLOUD	PRIVATE CLOUD	HYBRID CLOUD
SCALABILITY	High	Limited by in-house resources	High, leveraging both private and public resources
COST	Pay-as-you-go, generally lower upfront cost	Higher upfront cost for infrastructure	Variable, can optimize costs based on needs
CONTROL	Limited control over infrastructure	High control over resources and security	Balanced control
SECURITY	Standardized security managed by provider	Customizable, potentially more secure	Flexible, can be tailored to specific needs
COMPLIANCE	Depends on the provider	Easier to ensure compliance	Can be managed to meet specific requirements
BEST FOR	Startups, SMEs, non-sensitive data	Enterprises, sensitive data	Businesses with diverse and dynamic needs

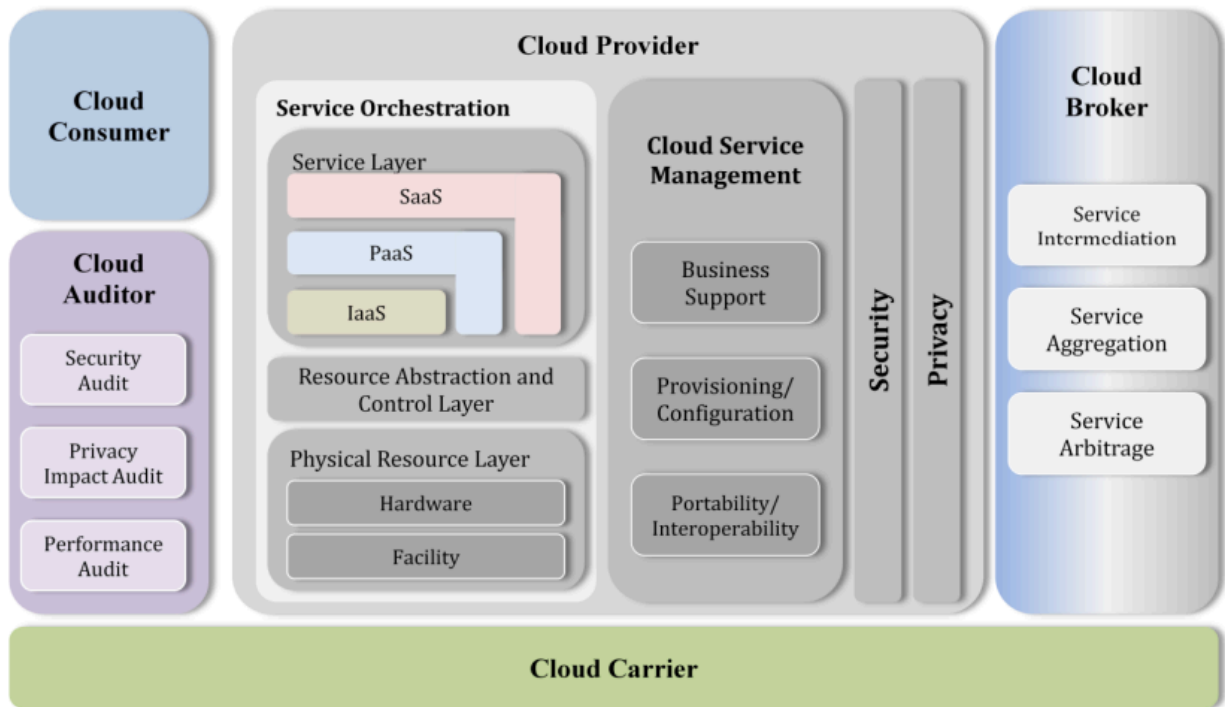


Figure 1: The Conceptual Reference Model

As shown, the NIST cloud computing reference architecture defines five major actors: cloud consumer, cloud provider, cloud carrier, cloud auditor and cloud broker. Each actor is an entity (a person or an organization) that participates in a transaction or process and/or performs tasks in cloud computing.

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, <i>Cloud Providers</i> .
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between <i>Cloud Providers</i> and <i>Cloud Consumers</i> .
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from <i>Cloud Providers</i> to <i>Cloud Consumers</i> .

Service Abstraction

The word 'abstraction' is derived from Latin words 'abs' meaning 'away from' and 'trahere' meaning 'to draw'. Abstraction takes away or hides the characteristics in order to reduce something into an essential utility. In computing, layer of abstraction is a way of hiding the implementation details of a particular set of functionality. In the cloud computing model, the level of service abstraction rises while moving from IaaS towards SaaS. At the IaaS level, consumers have the freedom of building computing infrastructure (essentially in virtual mode) from a set of available options. They can configure machines, setup networks and select storage. All of the devices are available in virtual mode and consumers should have the knowledge of building the required infrastructure from the scratch. At the PaaS level, the abstraction intensifies. The underlying virtual infrastructure of IaaS level remains hidden from the consumers. They can simply work with the chosen platform environment without knowing how the underlying system has been configured. At this level, consumers have full control over the PaaS environment and they can configure and manage the utilities like web servers, database servers, and different application development environments. This abstraction intensifies further at SaaS level where consumers remain unaware even about the technology of an application. They simply use applications without knowing anything about its implementation. Cloud computing abstracts the underlying computing system from the users. The degree of abstraction increases with the layers of services and abstraction is maximum at the SaaS level.

SPI Model

The three service models SaaS-PaaS-IaaS together are referred to as the SPI model of cloud computing. In this layered architecture of cloud service model, by moving upward, each service layer is empowered by the capabilities of the service layer(s) beneath it. Thus, PaaS is empowered by the IaaS layer and SaaS is empowered with the potentials of underlying IaaS and PaaS layers. One point to note here is that there is no functional relation between cloud deployment models and delivery models. Any delivery model can exist in any deployment scenario and thus any delivery/ deployment pairing is possible. SaaS offerings are mostly public services and hence the SaaS/public combination is more common than the other combinations.

Function	Cloud Computing	Traditional Computing
Business model	Pay for use (subscription model), administrative overhead is reduced	Pay for assets, administrative costs
Concept	Delivery of different services such as data and applications through internet on different servers	Delivery of different services on local server
Data access	Ability to access data anywhere at any time by end user	User can access data only on system in which it is stored
Costs	Most cost effective as operations and maintenance of server is shared among several parties which reduces cost of public services	Less cost effective as one has to buy expensive equipment to operate and maintain server
Connectivity	Requires fast, reliable, and stable Internet connection to access data and application	Does not require Internet connectivity to access data or application
Scalability and Elasticity	It is highly scalable and elastic one can increase or decrease computing resources as per business need	Scalability and elasticity are dependent on hardware, architecture and there is a limit to scope of expansion
Resiliency and redundancy	Resiliency and redundancy are built in by cloud providers	Resiliency and redundancy levels depend on architecture, additional costs are involved to build a high resiliency architecture

Scaling

Scaling is the characteristic of a system, model or function that describes its ability of growing or shrinking whenever required. In computing, scaling represents the capability of a system or application to deal with varying workload efficiently without bringing in a situation where resource shortage hampers performance or resource surplus increases the computation cost. In simple words, scaling is defined as the ability of being enlarged (or shrunk) for accommodating growth (or fall-off) to fulfill the business needs. A system or application architecture

can be termed as scalable if its performance improves on adding new resources and the improvement is proportional to the capacity added.

In a traditional computing environment, once the resource capacity of a system is enhanced manually the status of the system is retained (ever after) till further human intervention, even if those resources do not get utilized. This under-utilization causes wastage of resources and increases the cost of computing. Yet, very little can be done about it. The main reason behind this problem is that the infrastructure architecture is not dynamic in traditional computing systems, which prevents implementation of dynamic scaling. Static scaling requires system shut-down (system to restart) and hence is avoided unless it becomes extremely essential. For this reason, in a traditional static scaling environment, although resource capacity expansion was sometimes considered a possible option, capacity contraction was beyond imagination because service disruption had to be avoided.

Scaling in Cloud Computing

Scaling is one of the attractive attributes of cloud computing. Scaling in the cloud is dynamic in nature and apart from a few special cases it is automatic too. In dynamic-automatic scaling, the system resource capacity can be altered while a system is running. Large pool of virtualized resources promises to adjust variable workload by allowing optimum resource utilization. The scalability in cloud computing is offered in a transparent manner and the automatic scaling ability gives the impression of infinite resources to its consumers. Many cloud service providers even claim to offer infinite scalability although in reality that is not true. Even the largest players may at some moment face a scalability problem if cloud computing usage rate increases abruptly, beyond anticipation.

A computing system can be scaled by providing additional hardware resources. There are two different ways of supplying and integrating these hardware resources into a system. Depending on this, the scaling approaches can be categorized into two types as vertical scaling and horizontal scaling.

Vertical Scaling or Scaling Up

One way of increasing resource capacity is to replace the existing component (processors, memory etc.) with a more powerful hardware component. Thus, the capacity can be increased (or decreased) by upgrading (or reducing) a resource quality to support raising (or falling) workload. This type of scaling where a resource component is powered up through replacement is called scaling up or vertical scaling. For example, a system with dual-processor capacity can be scaled vertically by replacing the processor with a quad-processor. In cloud computing, increase in system capacity through vertical scaling is implemented by provisioning and integrating additional hardware resources within an existing resource node, instead of introducing new node to handle increased demand.

Advantages: It is not a complex task to replace a component of a system since it does not require alteration in the system architecture. So, the vertical scaling approach has low management complexity and is less risky.

Disadvantages: High power hardware components or specialized resources are expensive. Capacity or power of any resource component always has an upper limit. Hence, a sufficiently capable resource may not always be available or if available, the specialized hardware components may not be even affordable. On the other hand, the hardware replacement may cause interruption to the service and thereby introduce a system downtime problem.

Horizontal Scaling or Scaling Out

Resource capacity of a system can be increased by introducing additional resources (and without elimination of the existing). These new resources work together with the existing components.

This type of scaling where the system is equipped with supplementary resources to expand the resource capacity is known as scaling out or horizontal scaling. For example, a system with a dual-processor capacity can be scaled horizontally by adding another dual-processor. In horizontal scaling, the overall system capacity is enhanced by adding more nodes. In cloud computing, the increase in system capacity through horizontal scaling is implemented by provisioning additional resource nodes instead of replacing resource components of existing nodes with powerful components.

Advantages: It does not involve any system downtime. Specialized high-end hardware components are not required to scale out a system; generally available commodity hardware are used for horizontal scaling here. And more importantly, it is not limited by an individual component's capability as multiple similar type components (i.e. multiple processors or multiple memory units etc.) synchronize together in order to act as a powerful component. Unlike vertical scaling, no resource components are thrown out of the system (through replacement).

Disadvantages: It needs managing a large number of distributed nodes which need to synchronize together, and hence involve complexity. Presence of heterogeneous nodes further increases the complexity. The other issue is that horizontal scaling is appropriate for running distributed software applications; hence existing applications developed to run in a centralized environment are needed to be redesigned for a distributed computing environment.

<i>Vertical Scaling</i>	<i>Horizontal Scaling</i>
It is known as Scaling up.	It is known as Scaling out.
This approach involves replacement of components or resource nodes.	This approach involves introduction of additional components or resource nodes.
It can be implemented in any type of computing environment.	It can only be implemented in distributed computing environment.
It has less management complexity.	The management of larger numbers of nodes increases system complexity.
It may cause service interruption as system needs to restart after replacing component.	It does not cause service interruption. No system-restart is required.
It has less influence on application architecture.	This approach has a more fundamental influence on application architecture.
A vertically scalable application tends to run on high-end hardware.	A horizontally scalable application tends to run on low-end (commodity) hardware.
Here, the loads are concentrated.	It spreads the load.
It requires the specialized hardware components.	It can be done using normal commodity hardware components.
Upgrading the capacity of resource beyond a certain level can become very expensive in this type of scaling.	It is always less expensive and the expense is always proportional to resource capacity.
The expansion is limited by any resource's maximum capacity.	The expansion does not depend on available capacity of hardware components.
It is not a long term solution for scaling.	It provides a long term solution for scaling.

Although the architectural structures of the vertical and horizontal scaling approaches are different from one another, there are always trade-offs between the two models of scaling. The question raised here is whether it would be better to increase the power of the individual nodes or to combine the power of multiple nodes by distributing the load among them. The answer is not one. It differs on a case by case basis. The decision depends on two aspects: the computational and business requirements. From a computational viewpoint, the choice is primarily linked with factors like the architecture of the system and design architecture of application. From a business standpoint, the availability of requirements, cost of implementation and system performance are the decisive factors. In some cases, vertical scaling fits more appropriately and in many other cases horizontal scaling works well.

Horizontal scaling is a more Cloud Native approach:

Cloud computing model promises to deliver infinite scalability. Vertical scaling has limitations and it can only grow as per available resource components. Thus, the infinite scalability promise cannot be fulfilled with a vertical scaling approach in the cloud computing environment where application load may reach the zenith on different occasions. Horizontal scaling on the other hand can spread the load across multiple resource nodes and can thus support load unrestricted by any resource component's ability. The more resource nodes are added, the more load can be supported and thus this scaling approach is rather pertinent to deliver infinite scalability. The other issue is that in an auto-scaling environment, replacing a component with another is not a suitable choice. Rather, adding and releasing resources to scale is a much easier approach. Thus, the horizontal scaling approach becomes more cloud-native. But this does not entirely throw out vertical scaling from cloud computing, and in certain situations this scaling up process solves the issues faster.

Portability and Interoperability

The proliferation of cloud computing promises cost savings in technology infrastructure and faster software upgrades. The adoption of cloud computing depends greatly on how the cloud can address users' concerns on security, portability and interoperability. For portability, prospective customers are interested to know whether they can move their data or applications across multiple cloud environments at low cost and minimal disruption. From an interoperability perspective, users are concerned about the capability to communicate between or among multiple clouds. Cloud providers should provide mechanisms to support data portability, service interoperability, and system portability. Data portability is the ability of cloud consumers to copy data objects into or out of a cloud or to use a disk for bulk data transfer. Service interoperability is the ability of cloud consumers to use their data and services across multiple cloud providers with a unified management interface. System portability allows the migration of a fully-stopped virtual machine instance or a machine image from one provider to another provider, or migrate applications and services and their contents from one service provider to another. It should be noted that various cloud service models may have different requirements in related with portability and interoperability. For example, IaaS requires the ability to migrate the data and run the applications on a new cloud. Thus, it is necessary to capture virtual machine images and migrate to new cloud providers which may use different virtualization technologies. Any provider-specific extensions to the VM images need to be removed or recorded upon being ported. While for SaaS, the focus is on data portability, and thus it is essential to perform data extractions and backups in a standard format.

Security

It is critical to recognize that security is a cross-cutting aspect of the architecture that spans across all layers of the reference model, ranging from physical security to application security. Therefore, security in cloud computing architecture concerns is not solely under the purview of the Cloud Providers, but also Cloud Consumers and other relevant actors. Cloud-based systems still need to address security requirements such as authentication, authorization, availability, confidentiality, identity management, integrity, audit, security monitoring, incident response, and security policy management. While these security requirements are not new, we discuss cloud specific perspectives to help discuss, analyze and implement security in a cloud system.

Privacy

Cloud providers should protect the assured, proper, and consistent collection, processing, communication, use and disposition of personal information (PI) and personally identifiable information (PII) in the cloud.